

[Dashboard](#) / [My courses](#) / [COMS3005A-AAA3-S2-2024](#) / [Quizzes](#) / [Quiz 4 Greedy Algorithms, Dynamic Programming and Closest Pair](#)**Started on** Wednesday, 20 November 2024, 10:40 AM**State** Finished**Completed on** Wednesday, 20 November 2024, 11:04 AM**Time taken** 24 mins 13 secs**Grade** 12.67 out of 20.00 (63%)Question **1**

Correct

Mark 1.00 out of 1.00

Assume we implement a greedy approach to the making change problem in an attempt to make change for 15 cents. Select all of the coin sets below that will not result in an optimal solution when the greedy algorithm is used.

- ☒ a. 1, 2, 7, 10 ✓
- ☐ b. 1, 2, 5, 10, 25
- ☐ c. 1, 2, 5, 10, 15
- ☐ d. 1, 2, 5, 15, 25

Your answer is correct.

The correct answer is:

1, 2, 7, 10

Question **2**

Correct

Mark 1.00 out of 1.00

Consider "making change" problem, where we have a set of coin denominations and an amount that we need to make change for. The aim here is to determine the minimum number of coins that we can use to make up the required amount. We covered two approaches to this problem - a suboptimal greedy approach and an optimal dynamic programming approach. Given the coin denominations below, select the ones for which the greedy algorithm produces an answer (number of coins) that differs from the dynamic programming approach when the input amount is 62.

- ☐ a. 1, 7, 17, 20, 29, 32
- ☐ b. 1, 2, 5, 10, 20, 50
- ☒ c. 1, 3, 5, 25, 40 ✓
- ☒ d. 1, 2, 30, 40, 50 ✓
- ☐ e. 1, 2, 3, 4, 5
- ☐ f. 1, 4, 9, 17, 30

Your answer is correct.

The correct answers are:

1, 3, 5, 25, 40,

1, 2, 30, 40, 50

Question **3**

Correct

Mark 1.00 out of 1.00

Calculate the optimal service time for the following customers using the greedy approach.

Customer	Time
A	14
B	5
C	8
D	2
E	22
F	7

Answer:

81 ✓

The correct answer is: 81

Question **4**

Correct

Mark 1.00 out of 1.00

Selection Sort is a greedy algorithm.

Select one:

☒ True ✓

☐ False

always picks the smallest element and puts into the growing sorted list.

always picks the smallest element and puts into the growing sorted list.

The correct answer is 'True'.

Question **5**

Partially correct

Mark 2.00 out of 3.00

The Dynamic Programming change making algorithm is run to make change for 17 from a coin set of {1, 2, 5, 10}.

The table produced is

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_1 = 1$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_2 = 2$	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
$d_3 = 5$	0	1	1	2	2	1	2	a)	3	3	2	3	3	b)	4	3	4	4
$d_4 = 10$	0	1	1	2	2	1	2	2	3	3	1	2	2	3	3	2	c)	3

Give the missing answers

a) ✓

b) ✗

c) ✓

Your answer is partially correct.

You have correctly selected 2.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_1 = 1$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_2 = 2$	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
$d_3 = 5$	0	1	1	2	2	1	2	2	3	3	2	3	3	4	4	3	4	4
$d_4 = 10$	0	1	1	2	2	1	2	2	3	3	1	2	2	3	3	2	3	3

The correct answer is:

The Dynamic Programming change making algorithm is run to make change for 17 from a coin set of {1, 2, 5, 10}.

The table produced is

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_1 = 1$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_2 = 2$	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
$d_3 = 5$	0	1	1	2	2	1	2	a)	3	3	2	3	3	b)	4	3	4	4
$d_4 = 10$	0	1	1	2	2	1	2	2	3	3	1	2	2	3	3	2	c)	3

Give the missing answers

a) [2]

b) [4]

c) [3]

Question 6

Correct

Mark 2.00 out of 2.00

The Dynamic Programming change making algorithm is run to make change for 17 from a coin set of {1, 2, 5, 10}.

The table produced is

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_1 = 1$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_2 = 2$	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
$d_3 = 5$	0	1	1	2	2	1	2	a)	3	3	2	3	3	b)	4	3	4	4
$d_4 = 10$	0	1	1	2	2	1	2	2	3	3	1	2	2	3	3	2	c)	3

Condition $c[i, j] = c[i - 1, j]$ ✓ is applied to calculate the value in position $c[4,2]$ (change for 2 using 1, 2, 5, 10)

Condition $c[i, j] = 1 + c[i, j - d[i]]$ ✓ is applied to calculate the value in position $c[3,6]$ (change for 6 using 1,2,5)

Your answer is correct.

The correct answer is:

The Dynamic Programming change making algorithm is run to make change for 17 from a coin set of {1, 2, 5, 10}.

The table produced is

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_1 = 1$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
$d_2 = 2$	0	1	1	2	2	3	3	4	4	5	5	6	6	7	7	8	8	9
$d_3 = 5$	0	1	1	2	2	1	2	a)	3	3	2	3	3	b)	4	3	4	4
$d_4 = 10$	0	1	1	2	2	1	2	2	3	3	1	2	2	3	3	2	c)	3

Condition $c[i, j] = c[i - 1, j]$ is applied to calculate the value in position $c[4,2]$ (change for 2 using 1, 2, 5, 10)

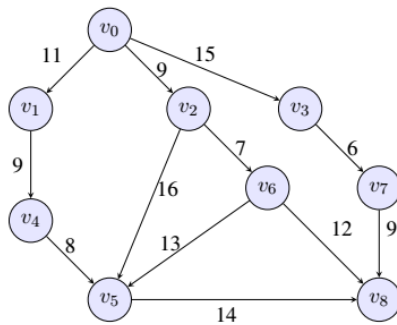
Condition $c[i, j] = 1 + c[i, j - d[i]]$ is applied to calculate the value in position $c[3,6]$ (change for 6 using 1,2,5)

Question 7

Partially correct

Mark 2.00 out of 4.00

Consider the graph shown below.



If the Dynamic Programming Weary Traveller algorithm is applied to this graph to calculate the shortest path from v_0 to v_8 then:

After the first pass through the While loop then D contains ✓ (where 99999 means infinity).

There are ✗ passes through the While loop.

The value of the shortest path once the algorithm terminates is ✓ .

The last node (before the start node) to have its distance to the goal node calculated is ✗ .

Your answer is partially correct.

You have correctly selected 2.

D is [99999, 99999, 99999, 99999, 99999, 99999, 99999, 99999, 0]

P is [1, 1, 1, 1, 1, 1, 1, 1, 0]

T is [0, 0, 0, 0, 0, 0, 0, 0, 1]

W is

[99999, 11, 9, 15, 99999, 99999, 99999, 99999, 99999]

[99999, 99999, 99999, 99999, 9, 99999, 99999, 99999, 99999]

[99999, 99999, 99999, 99999, 99999, 16, 7, 99999, 99999]

[99999, 99999, 99999, 99999, 99999, 99999, 6, 99999, 99999]

[99999, 99999, 99999, 99999, 99999, 8, 99999, 99999, 99999]

[99999, 99999, 99999, 99999, 99999, 99999, 99999, 99999, 14]

[99999, 99999, 99999, 99999, 99999, 13, 99999, 99999, 12]

[99999, 99999, 99999, 99999, 99999, 99999, 99999, 99999, 9]

[99999, 99999, 99999, 99999, 99999, 99999, 99999, 99999, 99999]

Pass 1 through the while loop

looping to consider all paths

considering node 0

node is in possible set

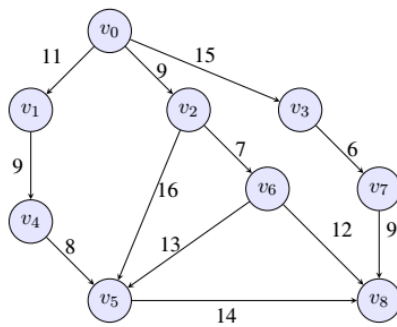
considering node 1

```
node is in possible set
considering node 2
node is in possible set
considering node 3
node is in possible set
considering node 4
node is in possible set
considering node 5
node is in possible set
considering node 6
node is in possible set
considering node 7
node is in possible set
considering node 8
D [99999, 99999, 99999, 99999, 99999, 14, 12, 9, 0]
T [0, 0, 0, 0, 0, 1, 1, 1, 1]
P [1, 1, 1, 1, 1, 0, 0, 0, 0]
Pass 2 through the while loop
looping to consider all paths
considering node 0
node is in possible set
considering node 1
node is in possible set
considering node 2
node is in possible set
considering node 3
node is in possible set
considering node 4
node is in possible set
considering node 5
considering node 6
considering node 7
considering node 8
D [99999, 99999, 19, 15, 22, 14, 12, 9, 0]
T [0, 0, 1, 1, 1, 1, 1, 1, 1]
P [1, 1, 0, 0, 0, 0, 0, 0, 0]
Pass 3 through the while loop
looping to consider all paths
considering node 0
node is in possible set
considering node 1
node is in possible set
considering node 2
considering node 3
considering node 4
considering node 5
considering node 6
considering node 7
considering node 8
D [99999, 31, 19, 15, 22, 14, 12, 9, 0]
T [0, 1, 1, 1, 1, 1, 1, 1, 1]
P [1, 0, 0, 0, 0, 0, 0, 0, 0]
Pass 4 through the while loop
looping to consider all paths
considering node 0
node is in possible set
considering node 1
considering node 2
considering node 3
considering node 4
```

considering node 5
considering node 6
considering node 7
considering node 8
D [28, 31, 19, 15, 22, 14, 12, 9, 0]
T [1, 1, 1, 1, 1, 1, 1, 1, 1]
P [0, 0, 0, 0, 0, 0, 0, 0, 0]
28

The correct answer is:

Consider the graph shown below.



If the Dynamic Programming Weary Traveller algorithm is applied to this graph to calculate the shortest path from v_0 to v_8 then:

After the first pass through the While loop then D contains [[99999, 99999, 99999, 99999, 99999, 14, 12, 9, 0]] (where 99999 means infinity).

There are [4] passes through the While loop.

The value of the shortest path once the algorithm terminates is [28].

The last node (before the start node) to have its distance to the goal node calculated is [v1].

Question 8

Partially correct

Mark 0.67 out of 2.00

Which of the following statements are true with regard to the dynamic programming approach to determining the most efficient order for performing matrix multiplications?

- ☐ a. The algorithm returns only the minimum number of multiplications required
- ☒ b. The algorithm returns the minimum number of multiplications required and the order in which the multiplications should be done. ✖
- ☐ c. The algorithm returns the order in which the multiplications should be done
- ☒ d. The cells below the main diagonal are not used in the calculations. ✔
- ☒ e. The process starts by setting all on the cells on the main diagonal to have the value 0. ✔

Your answer is partially correct.

You have correctly selected 2.

The correct answers are:

The algorithm returns only the minimum number of multiplications required,

The cells below the main diagonal are not used in the calculations.,

The process starts by setting all on the cells on the main diagonal to have the value 0.

Question 9

Incorrect

Mark 0.00 out of 1.00

Assume we are tackling a problem that is known to have overlapping subproblems. Select the statement below that is true

- ☐ a. We need more information before we can determine whether a dynamic programming approach will produce the optimal solution
- ☒ b. We can guarantee that a dynamic programming approach will produce the optimal solution ✖
- ☐ c. We can guarantee that a dynamic programming approach will not produce the optimal solution

Your answer is incorrect.

The correct answer is:

We need more information before we can determine whether a dynamic programming approach will produce the optimal solution

Question 10

Correct

Mark 1.00 out of 1.00

Mark which of the following statements apply to dynamic programming algorithms.

- ☐ a. They make locally optimal choices and hope these lead to globally optimal solutions
- ☒ b. They trade off storage space in order to increase speed of processing ✓
- ☒ c. They start with a top down approach to define a solution to the problem ✓
- ☒ d. They work bottom up to solve the problem ✓
- ☐ e. They always stick with the first solution that they find to a subproblem

Your answer is correct.

The correct answers are:

They trade off storage space in order to increase speed of processing,

They start with a top down approach to define a solution to the problem,

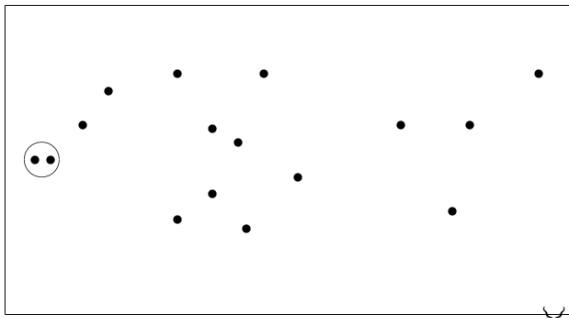
They work bottom up to solve the problem

Question 11

Correct

Mark 1.00 out of 1.00

Suppose the points shown in the figure below are the input to the final version of the closest pair algorithm.



How many times will the closest pair of points (indicated in the figure by the points with the circle around them) be involved in a direct distance calculation with each other?

Answer:



only at the limiting case of the recursion. One level up, one point will be outside the strip so won't be considered.

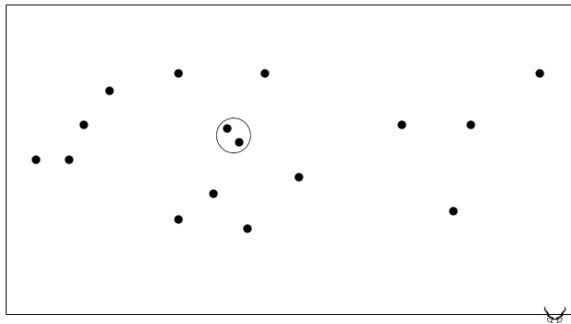
The correct answer is: 1

Question **12**

Incorrect

Mark 0.00 out of 1.00

Suppose the points shown in the figure below are the input to the final version of the closest pair algorithm.



Answer:

69



The correct answer is: 1

Question **13**

Incorrect

Mark 0.00 out of 1.00

Assume that we are attempting to compute the closest pair of points on a 2D plane, but we are guaranteed that all points are unique. We execute our $O(n \lg n)$ algorithm to solve this problem. When we check whether the closest pair of points lie in separate half-planes, how many neighbours must we now check?

Answer:

7



The correct answer is: 5

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